

Responsive layouts with Flexbox and Grid

Module 1 - Lesson 2

Overview

Flexbox and Grid are the two modern CSS layout engines. Flexbox lays out content along a single axis, while Grid gives you two-dimensional control over rows and columns. Knowing when to use each is the key to responsive design.

Key Concepts

- Flexbox arranges items along a main axis (row or column) and handles alignment and distribution with `justify-content` and `align-items`.
- Grid defines explicit rows and columns with `grid-template-columns` and `grid-template-rows`, then places items into cells.
- Use flex when the layout is a simple row or column; use grid when you need a full two-dimensional layout.
- Responsive breakpoints (`@media`) let layouts reflow at different viewport widths instead of squeezing.
- Modern techniques such as `minmax()`, `auto-fit`, and `fr` units remove the need for many breakpoints.
- Common patterns: sticky headers (flex), card grids (grid), and content-plus-sidebar layouts (grid areas).

Example

```
.cards {
  display: grid;
  grid-template-columns: repeat(auto-fit, minmax(240px, 1fr));
  gap: 16px;
}
.navbar {
  display: flex;
  justify-content: space-between;
  align-items: center;
}
```

Summary

Flexbox excels at one-dimensional alignment while Grid handles complex two-dimensional layouts. Combined with `auto-fit/minmax`, you can build layouts that adapt fluidly to any screen size.

Practice Checklist

- Rebuild a simple navbar with Flexbox and center items on both axes.
- Create a responsive card grid that goes from one column on mobile to three on desktop.
- Use grid areas to arrange a header, sidebar, main, and footer without extra markup.